



NEW PRODUCTS

MAKE IN INDIA

COMPONENTS & CHIPS

Contactless connectivity for point-to-point communication

Molex LLC has introduced a series of contactless connectivity devices, MX60, to ease device pairing. These devices integrate compact mm-Wave RF transceivers and built-in antennas within a single package. Operating at data rates from 1 to 5.4Gbps on



the 60GHz band without interference from Wi-Fi or Bluetooth, they prioritise energy

efficiency and speed. This series is poised to offer increased flexibility in product design, device pairing, and communication reliability. Sealed and contact-free, these solutions mitigate risks associated with ingress and repetitive motion, including high-vibration and rotational products. Additionally, eliminating diagnostic ports on wirelessly charged devices can lead to cost savings in development and production. The series finds applications in various sectors, from smartphones, AR/VR glasses, tablets, and self-driving vehicles to video walls, industrial robotics, medical wearables, wireless docking stations, and devices subjected to challenging environmental conditions.

Molex LLC

<https://www.molex.com>

Double-ridge waveguide couplers for RF systems

Pasternack has launched double-ridge waveguide couplers featuring integrated coaxial connectors for transitions in diverse applications. The couplers

offer good voltage standing wave ratio (VSWR) performance, achieving ratios as low as 1.25:1 to ensure efficient power transmission and minimise signal reflection for optimal signal integrity.



Additionally, the line introduces new industry benchmarks for broadband performance and efficiency

standards with its broadwall and loop coupler designs. The integrated connector design incorporates SubMiniature version A (SMA), Type N, and 2.92mm connectors, ensuring compatibility across a range of systems. The couplers provide flexibility with coupling levels spanning 10, 30, 40, and 50dB, catering to the adaptable requirements of diverse applications. These high-performance couplers are designed to address the intricate challenges prevalent in today's communications technology landscape, offering a maximum power rating of 400W (CW) to deliver exceptional performance even in the most demanding conditions.

Pasternack

<http://in.pasternack.com>

MOSFET bridging the gap in high-voltage designs

Littelfuse has launched CPC3981Z, an 800V, 100mA, 45-ohm small power



N-channel depletion mode MOSFET. The SOT-223-2L package, distinguished by the

omission of a middle pin, extends the drain-gate pin spacing from 1.386 millimetres to over 4 millimetres, enhancing

suitability for higher voltage applications such as switch mode power supplies (SMPS) and power factor correction (PFC) startup circuits by providing a greater creepage distance. This design allows designers to avoid the use of costly conformal coatings or potting. The product, suitable for industrial, energy, telecommunications, and LED lighting applications, offers a significant 800V blocking voltage rating. Its applications range from normally-on switches, power supplies, and startup circuits to solid-state relays, current regulators, and constant current sources.

Littelfuse

<https://www.littelfuse.com>

Next-generation motor position sensing innovation

Renesas Electronics Corporation has introduced its inductive position sensor (IPS) technology for motor position sensor ICs in robotics, industrial, and medical sectors. Distinct from magnet sensors, IPS units function at 600k



RPM and can determine startup position, featuring a 2µs propagation delay and capabilities for auto-calibration and

linearisation. By employing the Vernier principle, these dual-coil sensors offer a 19-bit resolution and a 14-bit accuracy. They are designed to operate effectively in diverse conditions, including extreme temperatures, dust, moisture, vibrations, and electromagnetic interference (EMI), and are shielded from stray fields. These sensors require no maintenance and provide a different error margin than magnetic sensors, making them a viable option over optical or magnet-based encoders. Inductive position sensor technology is also compatible with common communica-